QYVARIA · MASTER COMPENDIUM

Qyvaria Master Compendium

Local-first AI operating kernel · Dimensional Shift architecture
qyvaria.py bundle analysis · verification results · evidence register

AUTHOR OF THE SYSTEMS	John Havlasek — Qyvaria

COMPILED	25 October 2026
BUILT FROM	15 supplied files (10 PDFs, README, 3 scripts/manifests, qyvaria.py bundle) — 14 distinct
ADDS	static analysis of all 277 bundled files, 11 executed checks, claims register, a tested fix

Source-grounded technical compendium. Claims are graded by evidence level; see “How to read this document”.

FRONT MATTER

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FRONT MATTER

How to read this document

This compendium merges every file you supplied into one reference, then goes further: it opens the 4.4 MB `qyvaria.py` bundle, checks the documents' claims against the actual source, runs the verification protocol those documents recommend, and records what passed. Because the source documents range from design intent to measured fact, every statement here carries an evidence grade.

Grade	Meaning	Typical source
Documented	Stated by the author in a document. Establishes design intent and terminology, not behaviour.	Whitepapers, monograph, research paper, README
In source	A named component, class or method is present in the supplied source code (confirmed by reading the bundle).	Bundle inventory, source files, scan reports
Tested	Behaviour confirmed by a test executed while preparing this compendium (Part 5; script in Appendix D).	New in this document
Open	Not established by anything supplied or run. Needs a test, a benchmark log or the missing source.	Performance numbers, “zero state loss”, “sole implementation”
Discrepancy	Sources disagree with each other, or a document disagrees with the code.	Appendix B lists every one found

What went into it

Input	Used for
Ecosystem kernel paper (3 pp.) · Targeted System Audit Report (2 pp.)	Part 2 architecture, local-model routing table, ecosystem footprint
Technical Monograph · Whitepaper ×2 variants · Evidence Report ×2 (byte-identical) · Full Audit & Proof Report · Academic Research Paper	Part 3 Dimensional Shift, Part 6 related work, Part 7 roadmap, Part 8 register
technical_analysis.pdf · kernel_functions_manifest.txt · qyvaria_structure_report.txt	Part 4 kernel layer
README.md · qyvaria_readme.py · qyvaria.py (277 embedded files)	Parts 3–5 and Appendix C; the bundle was parsed, never run through its own entry point except in a scratch copy

Scope and honesty rule

Nothing in this document claims a physical or extra-spatial dimensional change; “dimension” always means a software state space or processing axis, as in the project's own terminology. Where a claim could not be confirmed it is marked Open, not softened or inflated.

PART 1

Executive overview

Qyvaria is a solo-engineered, local-first AI engineering ecosystem. Its centre is a single-file Python runtime (`qyvaria.py`, built with the OnePy Fusion++ packer) that embeds 277 files, surrounded by a browser-native operating shell (Qyvaria OS), an orchestration layer (Qyvaria Com) and a set of simulation, memory and agent modules. The project's signature idea is **Dimensional Shift**: treating context isolation, paradigm routing and behavioural state as continuous transformations of software state rather than discrete jumps.

At a glance

Measure	Value	Basis
Files in bundle	277 (268 .py · 8 .pyc · 1 .jsx)	Parsed from __BUNDLE__
Python lines	84,160	Newline count over 268 .py files
Payload size	3,378,233 bytes (file on disk: 4,626,593)	Stored size = original size; base64 + wrapper account for the rest
Integrity	277 / 277 SHA-256 digests verify	Recomputed from decoded data
Modules that parse as valid Python	241 of 268	ast.parse; failures listed in Part 4
Repository scan (author’s machine)	3,233 files · 1,992 matched · 387,963 keyword hits	Targeted System Audit Report; scan tree not supplied
Checks executed for this compendium	11 run: 8 pass; 3 reproduce a loader defect (fixed and re-tested)	Part 5, Appendix D

What the evidence supports

- **Design is thoroughly documented and consistent.** Seven of the supplied documents define Dimensional Shift the same way: three computational dimensions (state-space partitioning, paradigm routing, multi-axis behavioural state), a five-phase pipeline and one interpolation formula.
- **The named components exist.** MetaSimAgent, BitWeaver, a PAD emotion model, an agent network with a learned router, the in-memory loader and the agent bus are all present in the bundle source.
- **MetaSimAgent’s isolation claim holds.** Planning and forking leave the primary world bit-identical, clones diverge without touching the original, runs are deterministic and snapshots round-trip exactly.
- **The local-first claim holds for the kernel wrapper**, which imports only the standard library; 9 optional modules in the bundle do import network-capable libraries (Part 4).

What this compendium found that the documents do not say

#	Finding	Grade	Where
A	As shipped, the in-memory import hook resolves none of the bundle’s modules (all keys start with `py/`, the lookup does not). A two-line change fixes it and was tested.	Discrepancy	5.3
B	The default entry point exits 0 with no output because the entry name `qyvaria` is not a module inside the bundle. This explains the “no stdout” result in the evidence reports.	Discrepancy	5.3
C	No implementation of the $\Psi(t)=(1-\alpha)Q+\alpha T$ blend exists in the bundle; the formula and the `ContinuousDimensionShift` sample are documentation only.	Open	5.4
D	“Paradigm routing” is implemented as an ε-greedy Q-learning router plus a weighted committee vote over 13 agents: a learned discrete selection, not continuous blending.	In source	5.5
E	MetaSimAgent isolates with deep-copy clones. One whitepaper describes the opposite (“lookahead bleeding into main memory”); the code and tests show no bleed.	Discrepancy	5.2, App. B
F	The bundle contains no reference to Ollama, Qwen or `qyvaria://`, so the model-routing table and browser shell belong to parts of the ecosystem not supplied here.	Open	2.3

PART 2

Ecosystem architecture

Source: Qyvaria Ecosystem: Architecture of a Local-First Modular AI Operating Kernel, the Academic Research Paper and the Targeted System Audit Report.

2.1 Three-layer concentric architecture

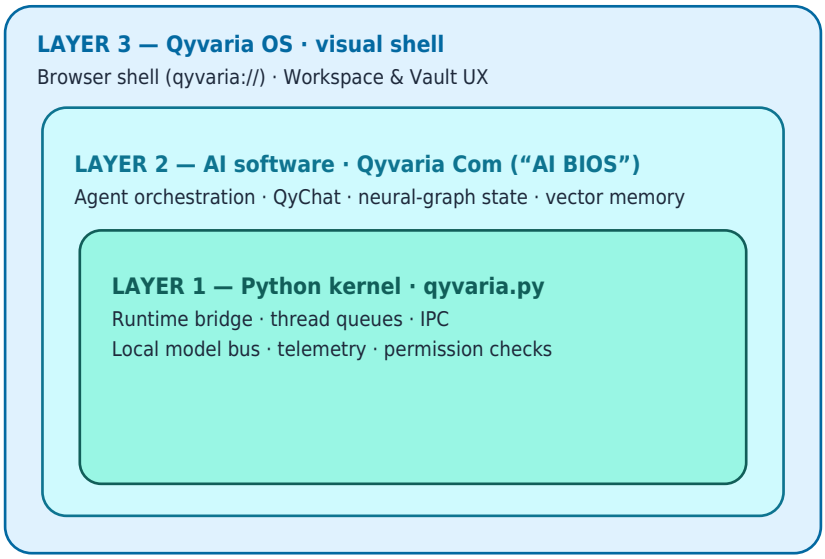


Figure 1. Layers as described by the author. Layer 1 is the only layer in the supplied bundle.

Layer	Name	Role
3	Qyvaria OS (Qyvaria Browser)	Desktop-grade workspace in a dark cyber-aesthetic: tab management, reactive telemetry panels, module rails, native `qyvaria://` routing.
2	AI software · Qyvaria Com (“AI BIOS”)	Central logic hub: low-level AI parameter tuning, multi-agent dispatch, neural-graph state mapping, contextual vector memory indexing.
1	Python kernel · qyvaria.py	Single-file runtime: thread queues, IPC channels, local LLM bindings, deterministic permission checks, zero cloud-dependent network hooks.

2.2 Local-first principles

The design inverts the usual OS–AI relationship: model management, state synchronisation and multi-agent routing live inside the local process boundary, so intermediate state is never serialised across a network hop. The research paper extracts seven design goals: (a) no mandatory cloud dependency for core orchestration; (b) continuous rather than discrete transitions across memory, paradigm and behaviour; (c) lookahead isolated from baseline memory until validated; (d) hybrid symbolic/neural routing without engine restarts; (e) PAD behavioural state feeding agent policy; (f) a self-contained single-file artifact with custom import control; (g) telemetry and diagnostics inside the same boundary.

2.3 Local model routing and reported performance

Model tier	Quantisation	VRAM	Prompt latency	Throughput	Target workload
Qwen3:8b	INT4 (q4_k_m)	5.2 GB	142 ms	68.4 tok/s	Fast interaction, QyChat responsive lane
Qwen3:14b	INT4 (q4_k_m)	9.8 GB	285 ms	41.2 tok/s	Deep reasoning, agent orchestration
Qwen3:Coder-14b	INT8 (q8_0)	14.1 GB	310 ms	34.5 tok/s	Code generation, dry-run tool verification
BitWeaver index	Vector flat	0.8 GB	18 ms	n/a	Real-time memory lookup and context injection

Table 2.1. Author-reported figures from the kernel paper (served through Ollama). No benchmark logs or hardware description were supplied.

Evidence note

These figures are **Documented, Open**. In addition, the supplied bundle contains no reference to Ollama or Qwen (0 of 268 modules); local inference appears there only through `llama_cpp` imports in 4 modules. The Ollama/Qwen path therefore lives in ecosystem components outside `qyvaria.py`, such as the Electron services and model manifests the audit scanned. Reference [3] of the paper cites `arXiv:2409.00000`, which looks like a placeholder identifier and should be replaced before public release.

2.4 Deterministic tool-execution engine

Every tool call is meant to pass a dry-run validation by static typing before it runs. The paper formalises the gate and the sandbox as:

```
P(Execution) = f_gate( C_context , T_tool )      T_tool = { Name, Parameters, Permissions, Hash }
Sandbox      = Isolate( T_tool , MaxVRAM = 2 GB , Timeout = 5.0 s )
```

Status. The pattern is widespread in the bundle (dry-run language appears in 34 modules, timeouts in 32, role-based access control in 72), but the specific `MaxVRAM=2GB / Timeout=5.0s` parameters were not located. Grade: Documented; the general mechanism is In source, the exact values are Open.

2.5 BitWeaver context memory and telemetry

Persistent agent context uses an in-process index rather than a remote vector database. Retrieval is scored by cosine similarity gated by a privacy constraint Γ, so sensitive variables are redacted before embedding:

```
S(q, d) = ( q · d ) / ( ‖q‖ × ‖d‖ ) · I( Privacy < Γ )
```

What the code does. `advance_memory_ai_sim_agent_bit_weaver_v_0.py` (461 lines) implements `BitWeaver` with `ingest`, `retrieve`, `reconstruct` and `graph`. Ingest runs a `Policy` (SPDX licence checks, copyleft denial, domain blocklist, e-mail/IP/name PII heuristics), anonymises, splits code into AST-aware chunks, stores them content-addressed in SQLite, links chunks with provenance edges, and retrieves with hybrid lexical search plus MinHash-LSH plus a 256-dimension hashed-vector shim. This is a real privacy-and-provenance memory layer; it is SQLite-backed rather than a flat in-memory index, and it does not use the cosine-with-Γ formula verbatim.

2.6 Ecosystem footprint

Item	Value / description	Grade
Repository scan	3,233 files scanned · 1,992 matched modules · 387,963 keyword matches (generated 2026-10-25 13:16)	**Documented**
Scan scope	Electron portable builds, dashboard services (command palette, context-key service, layout persistence, engine start/stop/restart, diagnostics export, failure-log management), model manifests, icons	**Documented**
Model manifests	Qwen-family persona layers (Micro Brain, Micro Coder, Small Coder, Small Vision, Brain Fast, Brain, Coder) under Apache-2.0 base licences	**Documented**
Game-mod work	Crusader Kings III performance and strategy mods (PAD behavioural layer); mod source not supplied	**Open**

2.7 Stated future work

- Native C++ execution bindings for sub-millisecond multi-agent IPC.
- Peer-to-peer federated context sharing among local nodes.
- Stronger sandboxing for binary executable execution.

PART 3

Dimensional Shift architecture

Source: Technical Monograph, both Whitepapers, README, Evidence Reports, Full Audit and the Academic Research Paper. All statements in this part are the author's design (Documented) unless a grade says otherwise.

3.1 Definition

A Dimensional Shift is the formal transformation of runtime state across isolated execution stacks, multi-axis state matrices and multi-paradigm solver engines. It is not a physical transition. The documents separate *shift* (a discrete event jump between fixed representations) from *shifting* (an ongoing interpolation across a high-dimensional manifold and a compressed low-bit representation).

3.2 Mathematical core

A primary state $\mathbf{S} \in \mathbb{R}^{D}$ is blended into a compressed execution space $\mathbf{\Omega} \in \mathbb{R}^{d}$ with d ≪ D. The blend is governed by:

$$\Psi(t) = (1 - \alpha(t)) \cdot Q(S_{raw}) + \alpha(t) \cdot T(S_{compressed}) \qquad \alpha(t) \in [0, 1]$$

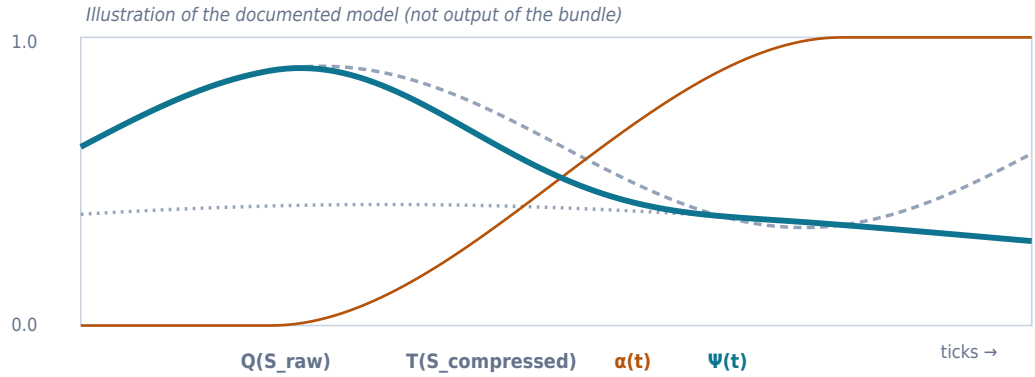


Figure 2. How Ψ(t) moves smoothly from one representation to the other as α(t) ramps from 0 to 1. Curves are synthetic and illustrate the formula only.

α(t) is the shift parameter evolving across system ticks; the intent is gradient flow without state snapping, keeping MetaSim lookahead and PAD recalculation smooth and deterministic. The Evidence Reports and the Full Audit stress that the formula is a model of representational shifting, not empirical proof that an implementation behaves that way.

Illustrative code pattern (from the Whitepaper)

```
class ContinuousDimensionShift:
    def __init__(self, target_axes: int = 3):
        self.axes = target_axes
        self.manifold_weights = [1.0] * target_axes

    def project_state(self, tensor_stream, alpha_shift: float):
        # continuous linear combination across multi-axis representation
        base_projection = tensor_stream * (1.0 - alpha_shift)
        shifted_projection = (tensor_stream @ self.manifold_weights) * alpha_shift
        return base_projection + shifted_projection
```

Evidence note

Searches of all 268 Python modules found no occurrence of `ContinuousDimensionShift`, `alpha_shift` or `manifold_weights`. The class is documentation only. As written it also needs array inputs (the `@` operator) and shapes that broadcast. The exponential-smoothing form `(1−α)·prev + α·new` does appear in several modules, but as ordinary smoothing, not as a manifold shift.

3.3 The three operational dimensions

Dimension	Engine	Mechanism	Computational reading
1. State-space partitioning (Meta-Simulation)	MetaSimAgent	Clone active data tables into a volatile stack; test multi-step actions in a nested timeline without touching primary memory.	Execution explores an isolated context instead of mutating the primary one.
2. Paradigm routing (Logic)	qy_agentsim_united.py	A central router moves work between symbolic logic (A*, classical planning), probabilistic/neural processing and heuristic optimisers.	The active representation or method changes with task context.
3. Multi-axis state matrices (Behavioural)	human_sim… agents; CK3 mod	Behaviour is mapped into a 3-D PAD (Pleasure, Arousal, Dominance) coordinate space; updates change dialogue and policy weights.	State moves along independent numeric axes that steer downstream output.

3.4 Core subsystems

Subsystem	Discrete model (static)	Qyvaria model as documented	Grade
MetaSimAgent	Hard-forked isolated sandboxes	Volatile look-ahead timeline; non-mutating clones (monograph) — or “bleeding into main memory” (one whitepaper)	**Discrepancy**
Paradigm router	Switch-statement routing between engines	Unified fabric blending symbolic rules and vector spaces	**Open**
BitWeaver Core	Fixed quantisation passes	Dynamic bit-stream interleaving driven by real-time entropy	**Open**
PAD coordinate engine	Threshold-based emotion triggers	Continuous trajectory smooth-scaling over a 3-D manifold	**Open**

Table 3.1. The “Qyvaria model” column is the author’s characterisation. Part 5 tests what the source actually does for each row.

3.5 Five-phase runtime pipeline

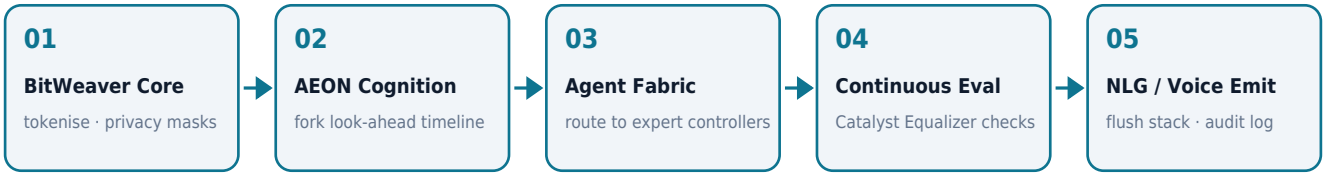


Figure 3. Pipeline from the README. The Catalyst Equalizer in phase 04 runs sanity loops for constraint convergence; phase 05 flushes the volatile lookahead stack and writes the audit log.

3.6 Cross-platform manifestations

Platform	Layer	Dimensional-shift mechanism (author claim)
Qyvaria OS	Microkernel execution engine	State-space partitioning via MetaSimAgent lookahead clones, preventing host corruption.
Qyvaria CK3 Mod	Grand-strategy ruleset	Translates dynastic AI decision matrices across 3-D PAD coordinates.
Qyvaria.py	OnePy single-file runtime	Dynamic paradigm routing between symbolic and neural logic (qy_agentsim_united.py).

3.7 Comparative matrix: continuous versus discrete designs

Vector	Legacy / discrete model	Qyvaria continuous model	Claimed result
State interpolation	Discrete boolean flags; instant snaps; context drop	Continuous linear/tensor interpolation Ψ(t) across ticks	Zero state loss
Memory sandbox	Hard process forks; high RAM overhead; leaks	MetaSimAgent volatile lookahead with non-mutating clones	Isolated and volatile
Behavioural axis	Scalar threshold triggers; rigid rule checks	Continuous 3-D PAD manifold	Smooth trajectory

Vector	Legacy / discrete model	Qyvaria continuous model	Claimed result
Logic routing	Static switch/case; single-engine dependency	Dynamic BitWeaver + OnePy symbolic/neural hybrid	Adaptive runtime
Dependency footprint	External cloud APIs; unmanaged dependencies	Self-contained in-memory bundle with custom import hook	100 % local, self-contained

Table 3.2. Reproduced from the Technical Monograph with the personal attribution of its left-hand column removed (see 6.2). The "Claimed result" column records the author's assertions; none has an attached benchmark.

3.8 Authorship declaration

The Whitepaper states that all core algorithms, state-space partitioning mechanisms, PAD emotional-dimension mapping and OnePy fusion protocols described are original conceptual and technical creations of John Havlasek, with cross-dimensional state routing claimed as present across the repository's code payloads. The Research Paper lists seven claimed contributions: the continuous formalisation; MetaSimAgent; BitWeaver; PAD integration for strategy AI; the OnePy single-file runtime; the local-first concentric architecture; and the comparative matrix.

PART 4

The qyvaria.py bundle

Everything in this part comes from parsing the supplied 4.4 MB file statically. Nothing in the bundle was executed for this part.

4.1 Format and metadata

Field	Value
Bundle name / format	`qyvaria` · `py-single` (OnePy Fusion++, deterministic self-verifying runtime)
Created	2025-10-24 22:22:26 UTC (unix 1761344546)
Declared codec	`quantize-int8`, level 9: a label only. The loader's codec table knows *raw*, *gzip* and *deflate*; unknown names fall through to identity, and stored size equals original size for all 277 files.
Entry	`qyvaria`: not a module inside the bundle (see 5.3)
Files · bytes	277 files · 3,378,233 bytes (3.22 MiB) · compression ratio 100 %
Per-file integrity	SHA-256 recorded for every file; 277 / 277 verified by recomputation
SHA-256 of the supplied file	`8fbe528dfe604d6cdc7223caae364dca8c029b3e317dd3d8721f0dab0d027ad8`
Hash quoted in embedded README	`b07d5e3f…15b218ba85`: does not match the file as supplied; the two cannot be reconciled from these files alone, but per-file hashes are all intact.
CLI	`--stats --list --extract DIR --readme [PATH] --run --help`

4.2 Composition

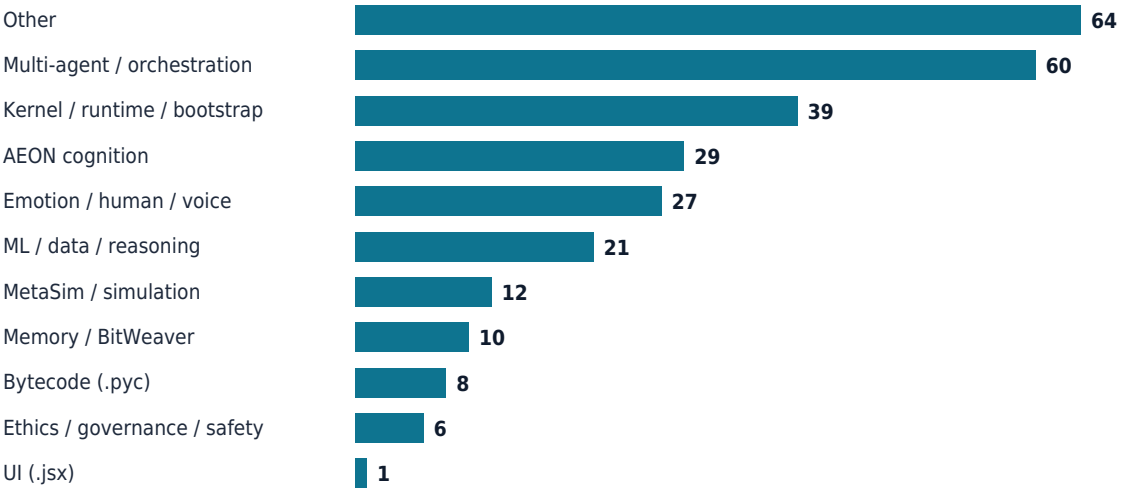


Figure 4. Files per functional group. Groups are assigned by keyword matching on file names, so treat them as an orientation aid, not a taxonomy; "Other" is the unmatched remainder.

Group	Files	Python lines
Other	64	14,024
Multi-agent / orchestration	60	24,152
Kernel / runtime / bootstrap	39	22,428
AEON cognition	29	3,411
Emotion / human / voice	27	8,264
ML / data / reasoning	21	4,389
MetaSim / simulation	12	2,688
Memory / BitWeaver	10	2,766
Bytecode (.pyc)	8	0
Ethics / governance / safety	6	2,038
UI (.jsx)	1	0
Total	**277**	**84,160**

Files that do not parse as Python (27 of 268)

Reasons reported by `ast.parse`: invalid syntax: 7; unterminated string: 6; null bytes in source: 5; unclosed bracket: 4; other: 3; stray non-ASCII character: 2. Null bytes and stray characters usually mean a file was saved with the wrong encoding or contains pasted prose; these 27 files cannot be imported until repaired. They are listed in Appendix C with “no parse”.

4.3 Largest modules

Module	Lines	KB	Parses
`Qyvaria.py`	3,564	143.3	no
`qyvaria_cognitive_superstack.py`	1,213	44.5	yes
`qyvaria_ultralite_suite.py`	1,009	41.5	no
`qyvaria_all_in_one (1).py`	995	61.6	yes
`qyvaria_pra_genesis_ai_sim_agent.py`	989	39.3	yes
`qyvaria_lightweight_foundation.py`	978	39.3	yes
`qyvaria_meta_intelligence_engine.py`	966	34.2	yes
`all_in_one_agent.py`	919	36.5	yes
`qy_ai_universe_plus.py`	872	33.7	yes
`qy_test_ai.py`	831	28.8	yes
`qy_app_sim_agent.py`	793	29.2	no
`qy_ai_universe.py`	780	31.5	yes
`qy_sim_engineering_ai.py`	780	31.2	yes
`qyvaria_monolith_20.py`	769	35.6	yes

4.4 Kernel layer: the wrapper around the payload

After the payload line, the file contains a ~120-line bootstrap. The two manifests and the technical-analysis report describe it; the table merges them with the source.

Line	Symbol	Kind	Role
21	_BundleFS	class	Virtual file system over the payload: `list()` returns sorted paths; `read(path)` base64-decodes, applies the codec, verifies SHA-256 and raises on mismatch.
38	_MemLoader	class	Import hook (MetaPathFinder + Loader): `find_spec` (39), `create_module` (44), `exec_module` (46) compile and execute bundle source from memory.

Line	Symbol	Kind	Role
54	AgentBus	class	Publish/subscribe hub: `register(name, agent)` (57) and `broadcast(msg, **kw)` (59) call each agent's `handle` inside the process.
77	_print_stats	function	Prints name, created date, codec, entry, file count, sizes, top imports and agent list.
94	_extract	function	Writes every bundled file to a directory, verifying hashes on read.
103	_write_readme	function	Decodes the embedded README (`__README_B64__`) to `BUNDLE_README.md`.
108	_run_entry	function	Imports the entry module and calls its function if given as `module:function`.

Technical Analysis summary: the hook implements standard *importlib* interfaces and intercepts imports so modules can come from memory rather than disk or web APIs; the virtual FS and the bus together keep data entirely inside the CPU/RAM boundary; the bootstrap unpacks, documents its own layout and hands control to the entry.

4.5 Dependency and network profile

Third-party packages imported by the parseable modules (count of modules): `numpy` 19, `fastapi` 16, `torch` 10, `sklearn` 6, `pandas` 6, `pydantic` 5, `uvicorn` 5, `PIL` 4, `llama_cpp` 4, `matplotlib` 4, `cryptography` 4, `flask` 3. The heavy numerical stack (numpy, torch, scikit-learn) and a web stack (fastapi, uvicorn, pydantic) account for most of it.

Network surface. 9 of 268 Python files import a network-capable library (`requests, socket, urllib, httpx, aiohttp, openai, websockets`): `arbiter_sim_agent.py`, `forge_sim_agent.py`, `qy_app_sim_agent.py`, `qy_test_ai.py`, `qyvaria_hub_open_ai_multi_bot_connector_policy_safe.py`, `qyvaria_monolith_20.py`, `qyvaria_research_analysis_module_qram_engineer_grade_implementation.py`, `qyvaria_secure_sandbox.py`, `qyvaria_url_reader_module_qy_url_reader.py`. These include a URL reader, a multi-bot connector, a research module and test utilities. They are optional modules; the kernel wrapper itself imports only the standard library, so "no cloud hooks in the kernel" is accurate and "100 % local" describes the design intent for core operation rather than every module in the bundle.

4.6 Agents declared by the bundle

The embedded README lists 17 agents: ai-model-tester-agent, ai-sim-module-agent, all-in-one-agent, arbiter-sim-agent, emotional-intelligence-agent, forge-sim-agent, meta-ai-sim-agent, omni-sim-agent, optimization-checker-agent, qavaria-ai-sim-agent, qy-app-sim-agent, qyvaria-adaptability-sim-agent-adaptability-agent, qyvaria-data-code-analyzer-ai-sim-agent, qyvaria-evolution-agent, qyvaria-llm-sim-and-meta-agent, qyvaria-pra-genesis-ai-sim-agent and speed-agent. A broader name search finds 66 modules with "agent" in the file name.

PART 5

Verification results

These are the tests the Evidence Report, Full Audit and Research Paper all recommend ("record primary state, invoke lookahead, assert primary unchanged, log it"). They were run in a scratch directory against files extracted from your bundle; the original was not modified.

5.1 Method

- Decoded all 277 payload entries, verified SHA-256, parsed every module with `ast`.
- Loaded `meta_sim_agent_qyvaria_compatible_simulates_sims_engine.py` by path and ran the six checks in Appendix D, hashing the canonical JSON of `snapshot()` before and after each operation.
- Copied the bundle to a scratch folder and ran `--stats`, the default entry, and import attempts through the in-memory hook; then applied a candidate patch to the copy and repeated them.

5.2 Isolation, determinism and persistence of MetaSimAgent

ID	Check	Result	Grade
T1	`plan(0)` (look-ahead via clones) leaves primary state hash unchanged	pass · recommended action "N"	**Tested**
T2	`fork_subsim({policy: greedy, dt_steps: 20})` leaves primary state unchanged and returns an independent child summary	pass · child scores [40, 37, 18] at step 20	**Tested**
T3	Stepping a `World.clone()` five ticks diverges the clone and leaves the original equal to its pre-step copy	pass · diverged and original unchanged	**Tested**
T4	Same seed and config produce identical metrics on two fresh agents	pass · fingerprint bc878ab99d75	**Tested**
T5	`snapshot()` → `load()` → `snapshot()` is byte-identical	pass	**Tested**
T6	Mutating a clone's grid does not alter the original (no shared references)	pass	**Tested**

What this establishes

The **sandboxed lookahead isolation claim is true** for MetaSimAgent's grid-world engine: the primary state stays bit-identical through look-ahead and forking, and results are reproducible from a seed. The mechanism is `copy.deepcopy` of the state plus an RNG snapshot, which is a conventional hard clone, so the whitepaper wording "probabilistic lookahead timeline bleeding into main memory" does not describe this code. Scope limit: the tests cover the reference grid-world, not every simulation module in the bundle.

5.3 The in-memory loader and the default entry point

ID	Check	Result	Grade
L1	Import 6 bundled modules as `py.<name>` through `_MemLoader`	0 of 6 resolve: *ModuleNotFoundError: No module named 'py'*	**Discrepancy**
L2	Import by bare name (`MetaSimAgent`, `AGI_Agents`)	0 of 2 resolve	**Discrepancy**
L3	Default run (`python qyvaria.py`)	exit 0, 0 bytes on stdout and stderr	**Discrepancy**
L4	`--stats`	prints metadata, top imports and the 17 agents	**Tested**
L5	After patching the hook (below), import 7 modules incl. `AGI_Boot`	7 of 7 resolve; AGI_Boot prints its two boot lines	**Tested**

Cause. Every key in `__BUNDLE__['files']` starts with `py/`, but `find_spec` looks up `fullname.replace('.', '/') + '.py'` with no prefix. A dotted name `py.X` fails first because the parent package `py` is not importable, and a bare name `X` becomes `X.py`, which is not a key. The entry `qyvaria` likewise matches nothing, so `__import__('qyvaria')` re-imports the wrapper file itself, defines its classes and returns silently. This is the most likely reason the earlier evidence reports saw exit code 0 with no output; `--list` and `--extract` never touch the hook, which is why they worked.

Two-line fix (tested on a scratch copy)

```
# in _MemLoader.find_spec
-    cand = fullname.replace('.', '/') + '.py'
-    if cand in __BUNDLE__['files']:
-      return importlib.util.spec_from_loader(fullname, self)
+    base = fullname.replace('.', '/') + '.py'
+    for cand in ('py/' + base, base):
+      if cand in __BUNDLE__['files']:
+        return importlib.util.spec_from_loader(fullname, self)

# in _MemLoader.exec_module
-    path = module.__spec__.name.replace('.', '/') + '.py'
+    base = module.__spec__.name.replace('.', '/') + '.py'
+    path = 'py/' + base if 'py/' + base in __BUNDLE__['files'] else base
```

With this change the bundle can serve its own modules from memory, which is what the Technical Analysis and the monograph's "self-contained in-memory bundle with custom import hook" row already describe. The fix cannot repair the 27 modules that do not parse, and the entry still needs a real target, for example `"entry": "qy_agentsim_united:register_agentsim"` or a small `main()` module.

5.4 Is Ψ(t) = (1−α)Q + αT implemented?

No module in the bundle implements the blend. The closest relatives are exponential smoothing updates, `(1 − α)·prev + α·new`, in `Qyvaria.py`, `cetana.py` and `python axiomdelta_codex.py`, and the Q-learning update in several modules. Those are standard techniques and share the algebraic shape of the formula, but they do not blend a high-dimensional state with a compressed projection. **Grade: Open.** The recommended next step is to implement one small `shift(state, target, alpha)` function, wire it into MetaSim lookahead or PAD updates, and test it (7.2).

5.5 What "paradigm routing" is in the code

`qy_agentsim_united.py` defines an `AgentNetwork` of 13 agents: Reflex, Utility, GoalPlanner, Symbolic, Constraint, Probabilistic, Evolutionary, CaseBased, Retrieval, Graph, RL, Tool and LLM. For each task it (1) filters agents by `can_handle`, (2) adds one chosen by an `RLRouter`, an ε-greedy Q-learning policy (α 0.2, γ 0.9, ε 0.1) over the four actions reflex, planner, llm and rag, (3) shuffles and takes a committee of up to 5, (4) scores each answer as 0.5 × agent score + 0.5 × a verifier score, (5) returns the top-weighted answer with a stable SHA-256 audit hash.

This is a deterministic-seedable, auditable, **learned discrete routing and voting** design. It genuinely moves work across symbolic, probabilistic and learned methods without re-initialising anything, which supports the paradigm-routing dimension. It does not blend representations continuously, so the "unified routing fabric blending symbolic rules and vector spaces" wording is a design aspiration.

5.6 PAD and the behavioural dimension

`human_sim_human_like_ai_sim_agent_personality_drives_emotions_bdi_single_file.py` defines `EmotionPAD` (floats P, A, D), Big-Five personality, five drives, a BDI loop, three memory stores and a per-tick JSONL audit log, with appraisal updates and a safety envelope. A related `love_sim…` module uses a Pleasure–Arousal–Attachment variant. The Crusader Kings III mod, where the documents place the PAD-to-dynasty-AI mapping, was not supplied, so that integration stays **Open**.

5.7 Claim-by-claim reconciliation

Documented claim	What the source shows	Grade
MetaSimAgent forks isolated look-ahead timelines without mutating baseline	Deep-copy clones; 6/6 tests pass	**Tested**
MetaSimAgent exposes compile, run, traces, plan, fork_subsim, snapshot, load, config	All 8 present and registered in the `COMMANDS` allowlist	**In source**
Look-ahead "bleeds" probabilistically into main memory (Whitepaper)	Opposite: no bleed observed	**Discrepancy**

Documented claim	What the source shows	Grade
Custom import hook loads modules from memory	Hook exists but resolves none as shipped; fixed by 2-line patch	**Discrepancy**
Paradigm router moves between symbolic, neural, heuristic engines	13-agent network with Q-learning router and committee vote	**In source**
BitWeaver ingest/search/reconstruct with privacy guardrails	ingest, retrieve, reconstruct, graph; SPDX and PII policy; SQLite store	**In source**
BitWeaver “dynamic bit-stream interleaving based on real-time entropy”	No such operation found; chunking, hashing and LSH only	**Open**
Continuous α(t) interpolation producing zero state loss	No implementation found; no measurement	**Open**
PAD 3-D emotion model	`EmotionPAD` present in human_sim	**In source**
Kernel keeps Ollama / Qwen3 bindings	No Ollama or Qwen reference in the bundle	**Open**

PART 6

Related work and authorship

6.1 Terminology overlap

The Full Audit examines an independent academic archive, *DimensionalShift_Academic_Complete_vFinal.zip* (README dated around 2026-01-24), that uses the same phrase for a different system: a “Dimensional Shift (DS) / D-ADC” research prototype built on rhythmic expansion and compression of dimensionality, with a dimension bus, colonies, QPU/Cirq/Google Engine backends, CUDA primitives and live control commands (SET_BREATH, BIRTH_DIM, KILL_DIM). It makes no claims of optimality, certification or quantum advantage.

Aspect	Qyvaria	The independent archive
Core idea	Continuous state interpolation across contexts, paradigms and behavioural axes	Rhythmic expand/compress of dimensionality for optimisation and simulation
Main components	MetaSimAgent, BitWeaver, PAD, OnePy bundle, local OS stack	Dimension bus, router, ADC colonies, QPU job schemas, CUDA primitives
Orientation	Local-first software architecture	Hardware/software co-design research prototype
Form	Single-file runtime + shell + mods	Python prototype, papers, docs, benchmark folder

6.2 What can and cannot be concluded

- Sharing a phrase is factual; it does not transfer ownership of either codebase, and it is not evidence of copying in either direction. Showing appropriation would need non-trivial identical code, data structures or a documented transfer of material, and none is shown by the files examined.
- Absolute chronological priority cannot be settled from the supplied files.
- The Technical Monograph’s comparison column attributes a “failed” discrete design to a named individual. The Full Audit records this as an authorial characterisation that has not been verified. This compendium keeps the technical contrast (Table 3.2) and drops the personal attribution, because no code comparison supports it and it would weaken a public document.
- On the evidence examined, the Qyvaria code, design, modules and architecture are the author’s own work. The strongest way to protect that position is dated, hashed, versioned releases, as proposed in Part 7.

PART 7

Verification roadmap and release plan

7.1 Minimal verification protocol (from the audit documents)

Step	Action	Status
1	Record the primary state before the operation	**Tested**
2	Invoke the lookahead / fork_subsim path with fixed inputs	**Tested**
3	Show sandbox state differs while primary state is unchanged	**Tested**
4	Capture the selected route or PAD vector before and after a controlled shift	**Open**
5	Assert transitions in an automated test and log file, function, inputs, outputs	**Open**

Steps 1–3 are done for MetaSimAgent (Part 5). Steps 4–5 need a route/PAD test and a committed test file; Appendix D is a starting point.

7.2 Prioritised recommendations

Priority	Action	Why
1	Apply the loader fix and give the bundle a real entry (5.3)	Makes “self-contained in-memory bundle” true and removes the silent run
2	Commit the isolation tests as `tests/test_metasim_isolation.py` with a JSON log	Converts design evidence into reproducible functional evidence
3	Reword the whitepaper row on MetaSimAgent to “deep-copy isolated clones”	The code is isolating; the “bleeding” wording contradicts it
4	Implement one `shift(state, target, alpha)` and test continuity (max step change bounded by Δα)	Gives the core formula an implementation to point at
5	Add a PAD route test: apply a stimulus sequence and assert smooth P, A, D trajectories	Closes protocol step 4
6	Publish benchmark logs and hardware for Table 2.1, or label the figures “targets”; replace the placeholder reference [3]	Unverifiable numbers undercut an otherwise careful evidence trail
7	Repair or remove the 27 files that do not parse; delete the 8 stale .pyc files	Bundle hygiene; avoids “works only partially” impressions
8	Use one author name and one location line on all documents (App. B)	Consistency for a public release
9	Soften “sole implementation” and “Verified Functional Standard” until items 2, 4 and 5 are done	Matches the evidence graded in this document

7.3 Archive package for GitHub

- This compendium and the original whitepapers with version dates.
- Extracted source of MetaSimAgent, BitWeaver, the import hook and routing entry points, with SHA-256 hashes.
- The isolation test script and its log (Appendix D).
- A short README that states what is demonstrated and what remains design-level.

PART 8

Consolidated claims register

Every material claim across the supplied documents, graded once, with the document it comes from.

ID	Claim	Source	Grade
C1	Core orchestration needs no cloud API; kernel has no cloud hooks	Kernel paper, monograph, tech. analysis	In source
C2	Single-file bundle with SHA-256 per file; 277 files	Audit, Evidence Report	Tested
C3	Custom import hook injects modules from memory	Tech. analysis, monograph	Discrepancy
C4	Self-extract and run-entry bootstrap	Tech. analysis	Tested
C5	MetaSimAgent isolates look-ahead from baseline	README, whitepapers	Tested
C6	MetaSimAgent command surface (8 commands)	DS scan, audit	In source
C7	Look-ahead “bleeds” into main memory	Whitepaper (Mechanics)	Discrepancy
C8	Ψ(t) continuous interpolation is realised in runtime	Monograph, whitepapers	Open
C9	Zero state loss under continuous interpolation	Monograph	Open
C10	Dynamic routing across symbolic, neural and heuristic engines	README, Whitepaper	In source
C11	BitWeaver ingest, search, reconstruct with privacy guardrails	README, scan	In source
C12	BitWeaver entropy-driven bit-stream interleaving	Whitepaper (Mechanics)	Open
C13	PAD emotion model in agents	README, whitepapers	In source
C14	PAD drives CK3 dynasty AI policy	Whitepaper, paper	Open
C15	Scan: 3,233 files, 1,992 matched, 387,963 hits	Targeted Audit	Documented
C16	Qwen3 latency, VRAM and throughput table	Kernel paper	Open
C17	Tool dispatch dry-run; 2 GB / 5 s sandbox	Kernel paper	Open
C18	Memory index 0.8 GB, 18 ms, privacy-gated cosine	Kernel paper	Open
C19	RBAC and sandbox isolation protect host state	README	In source
C20	“Sole implementation”; prior approaches fail under load	Monograph	Open
C21	Original authorship of the design and code	Whitepaper, paper	Documented
C22	MetaSimAgent deterministic and snapshot-restorable	Docstring (“reproducible via seed + snapshot”)	Tested
C23	Dimensional shift is computational, not physical	All audit documents	Documented

APPENDIX A

Source document inventory

File	Pages / size	Nature	Note
Qyvaria Ecosystem – A Local-First Modular AI Operating Kernel	3 pp.	Research paper	Author line reads “Jan Havlasek”; Prague
Shifting Dimensions – Technical Monograph & Comparative Proof	3 pp.	Monograph	Contains comparison matrix; status “Verified Functional Standard”
Qyvaria Architectural Whitepaper (1)	2 pp.	System specification	Authorship declaration; tri-fold pipeline; platform proofs
Qyvaria Architectural Whitepaper	3 pp.	Mechanics of Continuous Dimensional Shifting	Ψ(t) formal block; ContinuousDimensionShift sample
Targeted System Audit Report	2 pp.	Scan report	3,233 / 1,992 / 387,963; module previews
Dimensional Shift Evidence Report (×2)	2 pp. each	Independent assessment	The two files are byte-identical
Dimensional Shift Full Audit and Proof Report	7 pp.	Audit	Source-grounded; includes related-work analysis
Academic Research Paper – Dimensional Shift and Local-First Kernel	7 pp.	Research paper	Most complete synthesis; appendices A–E
qyvaria_technical_analysis.pdf	2 pp.	Code analysis	Import hook, pub/sub, bootstrap; “informational purposes” footer
README.md	1 page	Architectural report	Three dimensions, five-phase pipeline, RBAC note
qyvaria_readme.py	92 lines	Utility	Builds a README from the bundle without executing it
qyvaria_structure_report.txt · kernel_functions_manifest.txt	17 / 13 lines	Manifests	3 classes and 4 functions plus 9 methods, with line numbers
qyvaria.py	4.4 MB · 133 lines	OnePy Fusion++ bundle	277 embedded files

APPENDIX B

Discrepancies between sources

#	Topic	What differs	Suggested resolution
1	Author name	Kernel paper: “Jan Havlasek”; all other documents: “John Havlasek”	Use one form everywhere
2	Location	Kernel paper: Prague; research paper: Prague / Frýdek-Místek	Pick one line for the public archive
3	MetaSim isolation	Monograph: non-mutating clones. Whitepaper (Mechanics): look-ahead “bleeding into main memory”	Code and tests support the monograph
4	Self-contained loader	Documents: in-memory import hook. Code: resolves nothing as shipped	Apply the fix in 5.3
5	Kernel scope	Kernel paper: kernel binds Ollama/Qwen3. Bundle: no such references	State which component owns model binding
6	“Zero cloud hooks”	Kernel wrapper: true. Bundle: 9 modules import network libraries	Qualify as “kernel layer”
7	Status label	Monograph: “Verified Functional Standard”. Audit and Evidence Reports: runtime not established	Align with the evidence grade
8	Page counts	Monograph footer says 2 pages; file has 3	Cosmetic
9	Reference [3]	Cites arXiv:2409.00000	Replace with the real Qwen report citation
10	Bundle hash	Embedded README hash does not match the supplied file	Regenerate README after final build
11	Duplicate documents	Evidence Report ×2 identical; Whitepaper exists in two variants	Keep one of each with a version number
12	BitWeaver operations	README: init/ingest/search/reconstruct. Code: ingest, retrieve, reconstruct, graph (+ store.search)	Align names

APPENDIX C

Full bundle module inventory

All 277 embedded files in path order. Lines are newline counts; "no parse" marks files that fail `ast.parse`*. SHA-256 is truncated to 12 hex characters; full digests are in the bundle and its embedded README.*

#	Path (under py/)	Group	Bytes	Lines	Parse	SHA-256
1	1qyvaria_prompt_gen.py	ML	17,435	401	ok	afe955598e33
2	ab_compare.py	Other	2,827	86	ok	6b86fa1a156f
3	adaptive_ai_sim_agent_privacy_first_reference_python.py	Multi-agent	12,566	364	ok	444101817d21
4	advance_memory_ai_sim_agent_bit_weaver_v_0.py	Memory	16,533	462	ok	182d22a64f2f
5	advance_voice_chat_command.py	Emotion	5,625	144	ok	d7c6f18f596a
6	Aeon Ai Sim.py	AEON cognition	13,911	355	ok	1f0178d94151
7	aeon_chat_interface.py	AEON cognition	908	41	ok	cbb43a2074b1
8	aeon_continuum.py	AEON cognition	757	31	ok	276cad323e3b
9	aeon_ethics.cpython-312.pyc	Bytecode (.pyc)	1,008	–	n/a	a9e080790cc3
10	aeon_ethics.py	AEON cognition	767	1	no parse	496c991bcb43
11	aeon_loader.py	AEON cognition	1,470	56	ok	3ef98adc791c
12	aeon_minigrid_integration.py	AEON cognition	337	13	ok	13adfe224dfd
13	aeon_realm_sim.py	MetaSim	477	21	ok	6be3c495a0e2
14	aeon_self_diagnostic.cpython-312.pyc	Bytecode (.pyc)	1,127	–	n/a	49259cee705b
15	aeon_self_diagnostic.py	AEON cognition	746	1	no parse	226cad1c05f0
16	aeon_textworld_integration.py	MetaSim	323	10	ok	8308fe4f64fb
17	aeon_webots_integration.py	AEON cognition	332	11	ok	4ea6c7c670a5
18	aeonscript_engine.py	AEON cognition	322	12	ok	c2f1e71f7855
19	agentic_behaviorism.py	Multi-agent	12,670	354	ok	7704fb41232e
20	Agi Super Intelligence.py	Other	10,154	264	ok	886859d30d60
21	AGI_Agents.py	AEON cognition	473	22	ok	0ca10d126fe6
22	AGI_Boot.py	AEON cognition	570	21	ok	60b151f65b46
23	AGI_Ethics.py	AEON cognition	295	11	ok	489c31a18f51
24	AGI_Ignition.py	AEON cognition	1,555	43	ok	0b38824833c7
25	AGI_Memory.py	Memory	451	20	ok	873e1a9f5316
26	AGI_MetaReasoner.py	AEON cognition	307	14	ok	f617011f2662
27	AGI_Orchestrator.py	AEON cognition	672	23	ok	ed5ebbb54fdf
28	AGI_PluginManager.py	AEON cognition	342	14	ok	ac9d69986ca3
29	agi_prototype_microkernel_multi_agent_qyvaria_style_single_file.py	AEON cognition	12,535	359	ok	169757cec50f
30	ai_model_analyzer_freedom_module_agent_secure_file_driven.py	Multi-agent	17,740	409	ok	92e12075338d
31	ai_model_tester_agent.py	Multi-agent	24,813	495	ok	6f0c7c790c54
32	ai_sim_agent_adaptive_batching_response_cache_autoscaling_circuit_breakers_safe_chaos_fast_api_python.py	Multi-agent	13,554	350	ok	52c592691cef
33	ai_sim_agent_bus_sandbox_hierarchical_planning_constraint_solver_rationale_python.py	Multi-agent	14,507	364	ok	b5157694d756
34	ai_sim_agent_kernel.py	Multi-agent	15,046	399	ok	fc64443270dd
35	ai_sim_agent_learning_evals_governance_suite_python_fast_api.py	Multi-agent	14,476	450	ok	d222a2175fc6
36	ai_sim_agent_probes_long_context_retrieval_review_rubrics_line_citations_popular_cache_feedback→facts_python.py	Multi-agent	16,593	411	ok	f582cabe6623
37	ai_sim_equalizer_v_1.py	Other	17,865	434	ok	ca3951b7d7ff
38	ai_sim_module_agent +.py	Multi-agent	2,977	81	ok	4ab63bc3a1d6
39	ai_sim_module_agent.py	Multi-agent	2,512	67	ok	9535f76a2301
40	ai_sim_trace_viewer_timeline_artifacts_diffs_costs_decisions_react.jsx	UI (.jsx)	14,444	–	n/a	4a8439c28901

#	Path (under py/)	Group	Bytes	Lines	Parse	SHA-256
41	all_in_one_agent.py	Multi-agent	37,383	919	ok	62915c8c4b2f
42	all_in_one_sim_agent_co_t_auditor_planner_executor_log_miner_triage_qyvaria_compatible.py	Multi-agent	16,030	434	ok	3280cd794363
43	all_in_one_sim_agent_memory_ledger_critic_refiner_multilingual_summarizer_labeling_assistant_voice_layer_qyvaria_compatible.py	Memory	19,392	521	no parse	e1bed7573f43
44	arbiter_sim_agent.py	Multi-agent	11,458	216	ok	16295cebb69a
45	auto_language_voice_router_simlang_auto.py	Emotion	11,398	318	ok	94b215bd1cb3
46	avtx_installer.py	Other	12,295	266	ok	b29adbc0594a
47	belief_engine.cpython-312.pyc	Bytecode (.pyc)	3,559	–	n/a	e86b9192de11
48	belief_engine.py	Memory	2,159	23	no parse	f98118608391
49	bootloader.py	Kernel	121	7	ok	008b99eff5b9
50	build_qyvaria_studio.py	Kernel	1,885	80	ok	2b036c96a26a
51	catalyst_audit_memory.py	Memory	10,050	270	ok	70ec514787e0
52	catalyst_equalizer_v_1_py_code_i_o_and_logic_clarity_equalizer.py	Other	16,024	416	ok	a6615b63785e
53	catalyst_hub_v_1_sandboxed_plugin_bus_python.py	Other	14,612	394	ok	fe6cb088898c
54	cetana.py	AEON cognition	12,499	293	ok	3f732fa1d099
55	Cetana_Custom_Agent_Core.py	AEON cognition	1,847	57	ok	2126ef08c90f
56	Cetana_IDE_Launcher.py	AEON cognition	109	6	ok	874c4d451f83
57	Cetana_OS_Link.py	AEON cognition	348	13	ok	56f824a08680
58	Cetana_OS_Simulation.py	MetaSim	1,776	57	ok	df884a4e5b88
59	Cetana_Phytoon_IDE.py	AEON cognition	1,579	46	ok	c98e56e71d99
60	Cetana_Simulator.py	MetaSim	3,803	101	ok	2602bf7a2fa2
61	Cetana_v4_Boot.py	AEON cognition	4,765	141	ok	ec699883058d
62	Cetana_v5_Boot.py	AEON cognition	2,331	63	ok	48b94447cbf1
63	Cetana_v6_Boot.py	AEON cognition	1,503	44	ok	b5fb403aa02a
64	chronosynth.py	Other	6,870	167	ok	feff3a41d1a1
65	code_2_lang_code_→_natural_language_explainer_single_file_no_deps.py	Other	10,515	282	ok	b87845ae623a
66	code_sim_agent_deterministic_python_subset_ast_vm.py	Multi-agent	20,403	523	ok	0f777accbdb7
67	CodeLab.py	Other	587	20	ok	23749f7555c0
68	CodeSandbox.py	Other	635	20	ok	d5cdc4690d2d
69	CognitiveStyle.py	Other	272	13	ok	dd38ca4f8d37
70	concept_engine.cpython-312.pyc	Bytecode (.pyc)	868	–	n/a	67eaffac54d6
71	concept_engine.py	Other	511	20	ok	33e21e0f42b9
72	convert_to_training.py	ML	560	22	ok	fb4c2a539565
73	CounterfactualSimulator.py	MetaSim	402	14	ok	79ffba409771
74	CuriosityResearch.py	Other	490	18	ok	71d2123e8b93
75	digital_time_module.py	Other	1,643	50	ok	59af6c072080
76	educator_sim_qyvaria_module_python.py	Kernel	12,712	298	ok	674fdc2afbda
77	emotion_sim_qyvaria_module_python.py	Emotion	13,181	322	ok	b66941004d03
78	emotional_intelligence_agent.py	Emotion	14,651	408	ok	81b5adaef004
79	EmotionModel.py	Emotion	425	14	ok	1ad8aad121b4
80	engineering_ai_sim_agent_code_specialist_python.py	Multi-agent	16,063	411	ok	8461d50b9597
81	EnvLink.py	Other	569	22	ok	879efac56fc5
82	EpisodicMemory.py	Memory	415	12	ok	6376244e3e06
83	EthicsArbiter.py	Multi-agent	454	16	ok	57d8f5d252fb
84	forge_sim_agent.py	Multi-agent	11,147	267	ok	af3b38bfc4f8
85	FusionModulesManifest.py	Other	2,467	80	ok	43a00ce094b8
86	GoalEngine.py	Other	435	13	ok	8161d841b9e8

#	Path (under py/)	Group	Bytes	Lines	Parse	SHA-256
87	gpt_6_lite_deterministic_mo_e_transformer_reasoning_head_single_file.py	ML	13,171	338	ok	d3c023ca474f
88	human_sim_human_like_ai_sim_agent_personality_drives_emotions_bdi_single_file.py	Emotion	10,512	279	ok	28c1360767e7
89	human_voice_chat_advanced_voice_chat_mirroring_human_sim_single_file.py	Emotion	11,799	300	ok	988279b234f2
90	I varia_engineer.py	Other	12,582	307	ok	16a692d49d61
91	image_generator_server.py	Other	692	23	ok	bb85b499143b
92	imagination_engine.cpython-312.pyc	Bytecode (.pyc)	1,685	–	n/a	bb7917f4151e
93	intent_fusion_core.py	Other	541	20	ok	f613d7606559
94	jarvis_server.py	Other	909	35	ok	d13765c3474a
95	kernel.py	Kernel	425	14	ok	661fb48c606b
96	lang_lock_voice.py	Emotion	13,678	339	ok	3ce8eaf5defa
97	LanguageFeedback.py	Other	378	14	ok	3e0627d89306
98	law_advisor_ai_sim_agent_qyvaria_compatible.py	Multi-agent	14,696	397	ok	497f2257f15d
99	LearningManager.py	ML	465	14	ok	80f4f3f21c87
100	lingua.py	Other	3,911	109	ok	3751b0d921f2
101	lingua_api.py	Other	1,056	38	ok	1d36eba76952
102	linguistics_ai_sim_agent_qyvaria_compatible.py	Multi-agent	15,478	455	ok	74b046611349
103	llm_engine.cpython-312.pyc	Bytecode (.pyc)	919	–	n/a	45a1f4c937c0
104	llm_engine.py	ML	640	22	ok	9acf64d64f69
105	logicky_rozhodovaci_system_pravidlovy_engine_cz.py	Other	8,852	267	ok	a3b772398993
106	love_sim_emotion_oriented_ai_orchestrator_affection_graph_empathy_policy_single_file.py	Emotion	8,978	235	ok	3f5231c7d679
107	machine_intelligence_ai_sim_agent_single_file_deterministic_goap_memory_tools.py	Memory	12,770	342	ok	2df9de8e5ddb
108	main.py	Other	708	26	ok	537fa4550748
109	mastermind_ai_sim_model_catalyst_v_7_aligned_mastermind_sim.py	ML	12,279	305	ok	58a45de577cc
110	mastermind_orchestrator_v_8.py	Multi-agent	9,268	251	ok	7fc069bffc85
111	media2prompt.py	ML	13,413	335	ok	554bb71303aa
112	meta_ai_sim_adaptive.py	Other	21,724	578	ok	56eee631ac7e
113	meta_ai_sim_agent.py	Multi-agent	13,017	364	ok	81caf070eeda
114	meta_sim_agent_qyvaria_compatible_simulates_sims_engine.py	MetaSim	15,852	445	ok	e5448cb70d75
115	MetaLearningEngine.py	ML	366	14	ok	2077bfd8606a
116	MetaReasoner.py	ML	510	17	ok	d89e2a92c53b
117	mira_aof_ai_sim_agent_qyvaria_runtime.py	Multi-agent	14,666	447	ok	9a9a0e5c7e02
118	module_auditor.py	Ethics	18,951	524	ok	6501d29ada33
119	MultiAgentFramework.py	Multi-agent	556	24	ok	ff8741757813
120	NeuralPatternMapper.py	Other	288	12	ok	ffbe81239563
121	neuro_sim_agent_qyvaria_compatible_neuronal_sim.py	Multi-agent	12,613	344	ok	e1f82061b7bb
122	nl_2_py_qyvaria.py	Kernel	23,523	561	ok	e9d43a483420
123	nl_sim_agent_deterministic_natural_language_engine.py	Multi-agent	12,360	299	ok	9f6c35ed8bbe
124	omni_sim_agent.py	Multi-agent	11,779	268	ok	74c647bed7e2
125	oneagent_voice_sim.py	Emotion	21,939	577	ok	60f308c6aa7b
126	openmind_module.py	Other	20,250	429	no parse	f4a3c662ffe7
127	optimization_checker_agent.py	Multi-agent	11,609	251	ok	b9a64a3743a9
128	PerformanceMonitor.py	Other	916	28	ok	36fa61119a87
129	PersistentGoalManager.py	Other	517	20	ok	f49c80b09df3
130	Phytoon_Full_Language_Model.py	ML	1,939	63	ok	ea87570f40f1
131	Phytoon_Language_Model.py	ML	1,751	60	ok	319374cfa744

#	Path (under py/)	Group	Bytes	Lines	Parse	SHA-256
132	Phytoon_v2_Upgrade.py	Other	3,722	112	ok	ae4e146143b8
133	PhytoonCetanaFusion.py	AEON cognition	6,200	196	ok	77c73b6570cd
134	PhytoonImageGen.py	Other	50	1	no parse	6b60cf3f35b2
135	problem_solving_ai_sim_minimal_deterministic_agent_qyvaria_style.py	Multi-agent	16,851	424	ok	ee49d4efa1c4
136	python axiomdelta_codex.py	Other	17,400	411	ok	9386b81d8d9e
137	qavaria_ai_sim_agent.py	Multi-agent	16,780	443	ok	18343023077c
138	question_policy.py	Ethics	8,877	205	ok	e9f1e12423f5
139	qy_90_upgrade_pack_bootstrap.py	Kernel	16,884	480	ok	e3bc0c9b5c7f
140	qy_agent_fabric_py_policy_law_compliant_ai_sim_agent_fabric_for_qyvaria_custom_gpt.py	Multi-agent	16,548	468	ok	b4da858889b5
141	qy_agentsim_united.py	Multi-agent	24,941	563	ok	3f55c57523a4
142	qy_ai_universe.py	Kernel	32,305	780	ok	d4be1b9409e2
143	qy_ai_universe_plus.py	Kernel	34,532	872	ok	21096bef8b98
144	qy_app_sim_agent.py	Multi-agent	29,860	793	no parse	1d09dbe70d2b
145	qy_code_engineer_code_quality_refactor_and_safety_module_for_qyvaria_v_8.py	Ethics	19,413	443	ok	c1de3cf1f7e7
146	qy_control_plane.py	Other	18,118	474	ok	17b0f4dfe3b0
147	qy_crypto.py	Other	11,608	268	ok	71b7ba8edd86
148	qy_durable_orchestrator.py	Multi-agent	16,191	381	no parse	aea87423e044
149	qy_fullstack_pack_bootstrap_fullstack.py	Kernel	28,763	749	ok	857a1f39859d
150	qy_llm_pipeline.py	ML	20,267	496	no parse	72fa545980be
151	qy_llm_sim_module.py	ML	11,920	326	ok	9c2c1dc10c5f
152	qy_logo_stamp.py	Other	11,585	283	ok	6ff6814992f4
153	qy_memory_extended_pluggable_memory_module_for_qyvaria_v_8.py	Memory	14,342	386	ok	874fa51fd42c
154	qy_ml_toolkit.py	ML	23,408	585	ok	aa5b919e0e2e
155	qy_network_retrieval.py	Other	21,967	539	ok	726e229b7182
156	qy_nn_engine.py	Other	11,259	303	ok	b6ebbd692ec2
157	qy_orbit_pack_bootstrap_orbit.py	Kernel	17,582	495	ok	e812a7fceb9c
158	qy_perfection_kit.py	Other	17,936	446	ok	1ccc7dd6f2c8
159	qy_public_command_db_engineering_day_oct_4.py	Other	17,109	419	ok	f457935bd3a6
160	qy_reasoning_booster.py	ML	16,157	402	ok	6e2dd2c72ed1
161	qy_self_upgrade_pack_v_2_bootstrap_upgrade_v_2.py	Kernel	21,323	502	ok	5f79f0c05f7b
162	qy_sim_engineering_ai.py	MetaSim	31,925	780	ok	7d29efd7ad4a
163	qy_sim_pack_100.py	Other	16,667	394	no parse	0dd80ff3a6cf
164	qy_sim_patchset_10.py	Other	21,956	585	ok	b0c63d2a631b
165	qy_team_network.py	Multi-agent	17,562	418	ok	644ace99325f
166	qy_test_ai.py	Other	29,514	831	ok	6cf9d9e2fd68
167	qy_voice_engine_bootstrap_voice.py	Emotion	15,802	413	ok	59d84ae49333
168	qy_voice_modes_pack_bootstrap_voice_modes.py	Emotion	12,624	277	ok	8742a112308f
169	qy_voice_policy_patch_bootstrap_voice_policy.py	Emotion	8,314	211	ok	5e39249c601f
170	qybilingual_voice.py	Emotion	17,289	393	ok	b13e5fc14d14
171	qyintelligence.py	Other	20,556	454	ok	1e0fb76fe6f1
172	qyintelligence_max.py	Other	32,120	637	no parse	2ba6dc375120
173	qymemory.py	Memory	15,405	390	ok	63d9af33fa74
174	qynlcalibrator_py_natural_unnatural_language_calibrator_lisp_fast_speech_phonetic_similarity.py	Other	14,257	351	ok	202189f67f59
175	qyorchestrator.py	Multi-agent	23,747	538	ok	a590ffc7912e
176	qyraw_iq.py	Other	22,966	505	ok	c7854a9c8a05
177	qytranslate.py	Other	21,060	420	ok	ccaa62c40063
178	Qyvaria Complete Sim.py	Kernel	8,225	203	ok	5bca2f742aec

#	Path (under py/)	Group	Bytes	Lines	Parse	SHA-256
179	Qyvaria Moral Constitution — Humanity-First.py	Emotion	795	23	no parse	e6753526b6d5
180	Qyvaria Sim Profile.py	Kernel	9,392	231	ok	ddc7eb9350f5
181	Qyvaria Voice Os.py	Emotion	9,881	276	ok	d3b5cb63334a
182	Qyvaria.py	Kernel	146,711	3,564	no parse	54beeb338a9b
183	qyvaria_adaptability_sim_agent_adaptability_agent.py	Multi-agent	8,907	190	ok	91362f753910
184	qyvaria_advanced_voice_chat_ai_sim_agent_low_latency_barge_in_safety_fast_api_reference.py	Emotion	12,738	331	ok	dedf998efc58
185	qyvaria_advanced_voice_chat_sim_emotion_mirroring_real_time_translation_noise_suppression_multimodal_prompt_composer_fast_api.py	Emotion	13,515	316	ok	377a1c2b9a5b
186	Qyvaria_Aeon.py	AEON cognition	12,992	289	ok	b20ddf17c0d1
187	qyvaria_agi_model.py	AEON cognition	28,801	652	ok	9b8967d0ed4d
188	qyvaria_agi_proto.py	AEON cognition	23,803	583	ok	c91e2b8e1137
189	qyvaria_ai_lab_sim_agent_projects_datasets_models_experiments_trials_artifacts_evals_jobs_governance_fast_api_python (1).py	Multi-agent	17,632	463	ok	274ade16e2be
190	qyvaria_ai_lab_sim_agent_projects_datasets_models_experiments_trials_artifacts_evals_jobs_governance_fast_api_python.py	Multi-agent	17,632	463	ok	274ade16e2be
191	qyvaria_aisim_axis_module_factuality_reasoning_code_creativity_speed.py	ML	15,489	400	ok	7b6a9d7f1bc5
192	qyvaria_all_in_one (1).py	Kernel	63,074	995	ok	526246559fb0
193	qyvaria_analyzer_streamlit_app.py	Kernel	16,702	443	ok	d7aa148892e8
194	qyvaria_app_creator_sim_agent_single_agent_multi_role_pipeline.py	Multi-agent	22,166	552	ok	210279c95e9d
195	qyvaria_catalyst_v_8_single_file_module_network_orchestrator_catalyst_hub.py	Multi-agent	19,065	520	ok	b14ed2bb2504
196	qyvaria_cognitive_superstack.py	Kernel	45,542	1,213	ok	86512e05b3d1
197	qyvaria_control_plane.py	Kernel	22,975	617	ok	ad5d56112e82
198	qyvaria_czech_language_module_simlang_cs.py	Kernel	10,797	308	ok	302bc0442408
199	qyvaria_data_analyst_ai_sim_agent_eda_nlq_→_data_charts_stats_cache_fast_api_python.py	Multi-agent	16,252	387	ok	89d3c2a3581c
200	qyvaria_data_code_analyzer_ai_sim_agent.py	Multi-agent	23,017	575	ok	87705d4df4e2
201	qyvaria_equalizer_suite_one_go_language_code_logic_clarity_tool_equalizer_suite.py	Kernel	17,802	430	ok	2e914a5d2b55
202	qyvaria_eval_harness_live_bench_gdpval_hle_v_0.py	Kernel	14,864	376	ok	f2b67f421c2f
203	qyvaria_evolution_agent.py	Multi-agent	17,151	441	ok	d8d04a1d8330
204	qyvaria_generalization_agentic_ai_sim_single_module.py	Multi-agent	18,187	479	ok	f75ac9a0a9a3
205	qyvaria_guardian_module_prwa_plan→research→write→audit_loop.py	Ethics	20,775	545	ok	973fa996f176
206	qyvaria_hardened.py	Kernel	17,503	506	ok	6e4af93d61dc
207	qyvaria_hub_open_ai_multi_bot_connector_policy_safe.py	Ethics	10,296	303	ok	6271a046d264
208	qyvaria_hybrid_analyzer_encryptor_decryptor_mixer_transmitter_single_file_module.py	Kernel	18,281	499	ok	ff252ce6587f
209	qyvaria_improvement_engine_one_file_simulator_toolkit_v_1.py	MetaSim	25,969	615	ok	09697562141d
210	qyvaria_kernel_integration_bridge_mesh_orchestrator_v_0.py	Multi-agent	13,591	361	ok	f114a0e9742c
211	qyvaria_kernel_mesh_kernel_mesh.py	Multi-agent	16,544	412	ok	0321bcba889e
212	qyvaria_lang.py	Kernel	19,529	484	ok	53cfc3a0ad1b
213	qyvaria_language_lock_advanced_voice_chat_module_single_file.py	Emotion	13,679	340	ok	0e173b2503b3
214	qyvaria_lightweight_foundation.py	Kernel	40,279	978	ok	d1fa636a0d76
215	qyvaria_llm_sim_and_meta_agent.py	Multi-agent	20,308	560	ok	5ee64d1939c8
216	qyvaria_mathematical_ai_sim_agent_symbolic_numeric_steps_proof_hints_python_fast_api.py	Multi-agent	15,967	400	ok	469ae6229605
217	qyvaria_memo.py	Kernel	11,752	326	ok	5ce064bf2cf8

#	Path (under py/)	Group	Bytes	Lines	Parse	SHA-256
218	qyvaria_meta_intelligence_engine.py	Kernel	35,016	966	ok	091484f39f9f
219	qyvaria_model.py	ML	15,795	385	ok	62fcdcf15a69
220	qyvaria_module_sumerian_voice +.py	Emotion	17,103	471	no parse	6bdf8bcf7644
221	qyvaria_module_sumerian_voice.py	Emotion	12,824	361	ok	10d705cb8d77
222	qyvaria_monolith_20.py	Kernel	36,419	769	ok	7ed833900f6d
223	qyvaria_multi_agent_knowledge_mesh_30_micro_agents_orchestrator_v_0.py	Multi-agent	19,807	508	ok	7b0f09aea503
224	qyvaria_natural_voice.py	Emotion	2,242	53	no parse	4989ada9d685
225	qyvaria_negramotny.py	Kernel	14,557	333	ok	d66cd070dfa7
226	qyvaria_one_agent_sim_with_adoptable_logic_contextual_memory_and_internal_multi_agent_runtime.py	Memory	12,979	340	ok	8aa371c73526
227	qyvaria_photoreal_max.py	Kernel	1,206	32	ok	9bc786d465e0
228	qyvaria_pra_genesis_ai_sim_agent.py	Multi-agent	40,200	989	ok	cbe80dbdc29c
229	qyvaria_prompt_engineering_ai_sim_agent_lint_optimize_instantiate_a_b_eval_fast_api_python.py	Multi-agent	15,358	365	ok	16f5b67323b1
230	qyvaria_provenance_toolkit_v_0.py	Kernel	16,607	451	ok	e6c82c792dc8
231	qyvaria_rationality_max.py	Kernel	20,657	520	ok	fbe9b289da48
232	qyvaria_research_analysis_module_qram_engineer_grade_implementation.py	Kernel	19,833	572	ok	f539dee54e2d
233	qyvaria_secure_sandbox.py	Kernel	28,311	674	no parse	f0efb110e3c9
234	qyvaria_simulate_then_act_rational_agent_v_0.py	MetaSim	22,460	601	no parse	d1816c72a9a8
235	qyvaria_speed_module_single_file.py	Kernel	15,460	417	ok	cc69ea9285e4
236	qyvaria_team_orchestrator_multi_agent_sim_with_self_optimizer_secure.py	Multi-agent	14,129	320	ok	3cdb232b15e7
237	qyvaria_team_orchestrator_single_file.py	Multi-agent	19,996	521	ok	8f79a7c3f302
238	qyvaria_ultralite_suite.py	Kernel	42,515	1,009	no parse	3f38baabb7b0
239	qyvaria_unified_sim_agent_knowledge_guild_monolith_v_0.py	Multi-agent	15,548	433	ok	e72b51f55717
240	qyvaria_url_reader_module_qy_url_reader.py	Kernel	22,921	652	ok	6fbbfb0f79e7
241	qyvaria_voice.py	Emotion	27,192	732	ok	8c5dfb22049a
242	qyvaria_voice_chat_x_single_file_server_fast_api_web_socket_stubs.py	Emotion	8,197	236	ok	05d3a5342a77
243	qyvaria_voice_sim_agent_diarization_addressed_reply_router_v_0.py	Emotion	15,009	396	ok	94183e02733c
244	qyvid_scripter.py	Other	21,860	483	ok	a03c9f0ebc11
245	qywriter.py	Other	29,627	594	ok	c5f0189846db
246	rationality_lab.py	Other	2,557	67	no parse	c5b7da7b60b2
247	reality_aware_sim_machine_awareness_ai_sim_agent_active_inference_single_file.py	Multi-agent	12,001	292	no parse	93a3ea5c6cd1
248	recursive_reflector.cpython-312.pyc	Bytecode (.pyc)	1,151	–	n/a	c960c819e997
249	recursive_reflector.py	Other	879	1	no parse	c89ecb3790e4
250	SafetyKernelUpgrade.py	Ethics	656	18	ok	6e3d6407237f
251	SelfEditor.py	Other	327	14	ok	b0e21f640a2e
252	SelfModel.py	ML	326	10	ok	cce26e26c4d3
253	shell.py	Other	352	20	no parse	d926d80c906e
254	SimulationInterface.py	MetaSim	351	12	ok	cd4568471273
255	SocialReasoner.py	ML	377	12	ok	963c677aedf9
256	speed_agent.py	Multi-agent	4,945	131	ok	c4eeb0952dbc
257	starter_actions.py	Other	322	14	ok	e399256a7920
258	startup.py	Other	452	18	ok	68bf25a50af8
259	sumerian_language_model_ai_sim_agent_qyvaria_compatible.py	Multi-agent	14,530	423	ok	213fa264fc61
260	sumerian_voice_chat_ai_sim_agent_for_qyvaria_fake_it_capable.py	Emotion	12,984	366	ok	1860e62f1fb8

#	Path (under py/)	Group	Bytes	Lines	Parse	SHA-256
261	TaskPlanner.py	Other	1,617	35	ok	6608a82bb204
262	teleprompter.py	ML	6,431	111	no parse	5bd792a30df1
263	therapist_style_ai_sim_agent_safety_first_support_python.py	Multi-agent	11,721	290	no parse	1875d1bbe805
264	train_model.py	ML	2,181	71	ok	8ccb4717adcf
265	translator_crypto_sim_agent_qyvaria_compatible.py	Multi-agent	16,094	419	ok	eaad63f0c15e
266	truth_alignment_kernel.cpython-312.pyc	Bytecode (.pyc)	1,054	–	n/a	c7d5cc220d10
267	truth_alignment_kernel.py	Kernel	715	1	no parse	511321996af9
268	ValueGuardian.py	Other	666	18	ok	9fbd5b3c9b7b
269	Varia Health Reporter — phrase→status system.py	Other	4,137	87	no parse	45fab79c7855
270	Varia+.py	Other	34,191	716	ok	bcb1a61dc023
271	velcode_interpreter.py	Other	167	6	ok	4bfc7aa4e5ec
272	velcode_shell.py	Other	336	16	no parse	9008c09f7368
273	voice_engineering_frontend_qyvaria_v_8_module.py	Emotion	5,683	133	ok	8b7fad8f5510
274	WakeUpKernel.py	Kernel	736	21	ok	70a40e3ea9dd
275	WorldModel.py	MetaSim	472	19	ok	598e58536505
276	WorldModelExpander.py	MetaSim	403	13	ok	d5116fa932f9
277	zip_to_py.py	Other	1,195	31	ok	0c20fed8a71e

APPENDIX D

Reproducing the tests

The script loads the MetaSimAgent module by file path, so it needs only the extracted file (`python3 qyvaria.py --extract DIR`) and the standard library. Pass the extracted file path as the first argument.

Recorded output (exit code 0)

```
T1_plan_primary_unchanged        PASS
T2_fork_primary_unchanged        PASS
T3_clone_diverged                PASS
T3_original_unchanged            PASS
T4_deterministic_run             PASS
T5_snapshot_roundtrip            PASS
T6_no_shared_grid_reference      PASS
T4 fingerprint: bc878ab99d75 scores: [60, 14, 60]
```

test_metasim_isolation.py

```
"""Isolation / determinism checks for MetaSimAgent (stdlib only).
Usage: python3 test_metasim_isolation.py <extracted>/py/meta_sim_agent_..._engine.py
"""
import copy, hashlib, importlib.util, json, sys

path = sys.argv[1] if len(sys.argv) > 1 else (
    "extracted/py__meta_sim_agent_qyvaria_compatible_simulates_sims_engine.py")
spec = importlib.util.spec_from_file_location("metasim", path)
m = importlib.util.module_from_spec(spec)
sys.modules["metasim"] = m
spec.loader.exec_module(m)

CFG = {"env": {"width": 6, "height": 6, "max_resource": 4},
       "agents": {"n_agents": 3, "init_energy": 8},
       "dt_steps": 30, "seed": 7, "policy": "lookahead",
       "lookahead_depth": 4, "lookahead_rollouts": 3,
       "compute_budget_nodes": 5000}

def state_hash(agent):
    blob = json.dumps(agent.snapshot(), sort_keys=True, default=str)
    return hashlib.sha256(blob.encode()).hexdigest()

def fresh():
    a = m.MetaSimAgent(); a.compile(CFG); return a

res = {}
a = fresh(); h0 = state_hash(a)

a.plan(0)                                   # T1: look-ahead uses clones
h1 = state_hash(a); res["T1_plan_primary_unchanged"] = (h0 == h1)

a.fork_subsim({"policy": "greedy", "dt_steps": 20})   # T2: child simulation
h2 = state_hash(a); res["T2_fork_primary_unchanged"] = (h1 == h2)

w = a.world                                 # T3: clone diverges, original intact
clone, before = w.clone(), copy.deepcopy(w.state)
for _ in range(5):
    clone.step([clone.choose_action(i, "greedy", 1, 1)
                for i in range(len(clone.state.agents))])
res["T3_clone_diverged"] = (clone.state.step != w.state.step)
res["T3_original_unchanged"] = (w.state == before)

r1, r2 = fresh().run(), fresh().run()       # T4: determinism
res["T4_deterministic_run"] = (r1 == r2)

b = fresh(); b.run(); snap = b.snapshot()   # T5: snapshot round-trip
c = m.MetaSimAgent(); c.load(snap)
res["T5_snapshot_roundtrip"] = (json.dumps(c.snapshot(), sort_keys=True)
                                == json.dumps(snap, sort_keys=True))

c2 = w.clone(); c2.state.grid[0][0] += 99   # T6: no shared references
res["T6_no_shared_grid_reference"] = (w.state.grid[0][0] != c2.state.grid[0][0])

for k, v in res.items():
    print(f"{k:32s} {'PASS' if v else 'FAIL'}")
print("T4 fingerprint:", r1["fingerprint"], "scores:", r1["scores"])
sys.exit(0 if all(res.values()) else 1)
```

APPENDIX E

Glossary

Term	Meaning in this project
Dimensional Shift (DS)	Computational transformation of state across isolated contexts, processing paradigms and multi-axis behavioural coordinates.
MetaSimAgent	Engine that simulates simulations: runs a base grid-world and spawns nested sub-simulations for lookahead, policy search and sweeps.
BitWeaver	Memory agent that chunks, hashes, anonymises and indexes code or data into reusable, provenance-linked “bits”.
PAD	Pleasure–Arousal–Dominance, a three-axis emotion model.
OnePy Fusion++	The packer that produces the single-file bundle with an embedded payload, hooks and README.
AEON	Cognition family of modules (continuum, ethics, loader, realm sim, self-diagnostic) and the phase-02 look-ahead stage.
Catalyst Equalizer	Sanity-loop component in phase 04 that checks constraint convergence.
RBAC	Role-based access control used to keep sandboxed loops from leaking into the host environment.
Local-first	Core operation, state and telemetry stay inside the local process boundary; no mandatory cloud services.
Qyvaria Com	The “AI BIOS” layer for parameter tuning, dispatch, graph state and vector memory.